

CHESAPEAKE LIGHT, USA

WMO: 994020

Lat: 36.910N

Lon: 75.710W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (W05)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	-2.9	-13.7	1.1	-3.3	-11.7	1.4	-1.7	19.8	0.7	18.6	3.2	10.0	330	0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	29.1	25.7	28.1	25.5	27.4	25.3	27.5	28.1	26.8	27.5	26.2	26.9	6.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.4	23.2	27.8	26.7	22.3	27.2	26.1	21.4	26.6	87.2	28.1	84.3	27.6	81.4	27.0	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 22.1	(b) 18.3	(c) 15.8	(e) -7.6	(f) 31.4	(g) 3.2	(h) 0.9	(i) -9.9	(j) 32.1	(k) -11.8	(l) 32.6	(m) -13.6	(n) 33.1	(o) -16.0	(p) 33.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	5.2	5.4	8.4	13.0	17.4	22.5	25.1	24.8	22.5	18.0	12.5	8.6
	DBStd	7.87	4.84	4.08	3.87	3.55	3.24	2.76	1.67	1.70	2.07	3.12	3.81	4.12
	HDD10.0	483	161	137	78	12	0	0	0	0	0	1	19	75
	HDD18.3	1827	407	363	308	163	57	4	0	0	1	44	177	302
	CDD10.0	2429	12	8	28	102	229	376	467	460	375	247	94	31
	CDD18.3	731	0	0	0	3	28	130	209	202	126	33	2	0
	CDH23.3	3665	0	0	0	3	42	511	1412	1300	370	28	0	0
	CDH26.7	350	0	0	0	0	3	51	162	118	16	1	0	0
Wind	WSAvg	8.1	8.8	8.9	9.6	9.4	8.2	6.9	6.5	6.4	7.5	7.8	8.4	8.5
Precipitation	PrecAvg	1404	109	93	111	95	110	128	158	153	144	106	90	106
	PrecMax	1872	191	197	224	178	204	239	254	291	506	222	235	218
	PrecMin	979	45	36	32	34	42	46	93	60	36	3	21	43
	PrecStd	187	36	41	38	35	42	47	38	54	98	49	52	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	17.4	20.2	23.0	26.3	29.9	30.8	30.5	28.3	25.7	21.1	19.2
		MCWB	15.3	14.6	14.9	17.1	21.5	25.6	26.5	26.5	25.0	23.1	20.6	17.8
	2%	DB	14.9	14.7	17.4	20.6	24.5	28.2	29.2	28.8	27.0	24.0	19.9	17.3
		MCWB	13.9	12.7	15.5	16.9	22.0	25.3	26.2	25.3	25.0	22.4	19.1	16.5
	5%	DB	13.4	12.5	15.4	19.3	22.9	26.8	28.3	27.9	26.2	23.2	18.7	15.8
		MCWB	12.4	10.4	13.2	16.5	21.2	24.7	25.9	25.4	24.6	22.0	17.5	15.2
	10%	DB	11.9	10.7	13.9	17.8	21.7	25.8	27.5	27.2	25.4	22.1	17.5	14.3
		MCWB	10.8	9.1	12.1	15.9	20.6	24.0	25.5	25.2	23.6	20.3	16.2	13.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	15.6	17.6	19.5	24.0	28.2	28.0	28.9	27.4	24.4	20.5	19.1
		MCDB	16.6	16.3	18.1	20.4	24.5	28.6	28.6	29.6	27.7	24.6	20.7	19.2
	2%	WB	14.6	13.3	16.0	18.3	22.9	26.7	27.5	27.4	26.4	23.5	19.0	17.4
		MCDB	15.1	14.2	17.1	19.3	23.5	27.3	28.4	27.6	26.6	23.5	19.2	17.7
	5%	WB	12.9	11.2	14.5	17.4	21.9	25.8	26.9	26.6	25.6	22.7	17.4	15.9
		MCDB	13.4	12.3	15.9	18.7	22.6	26.6	27.8	27.0	26.1	22.8	17.7	16.1
	10%	WB	11.2	9.3	12.7	16.3	21.1	25.1	26.3	26.0	24.7	21.2	16.4	14.6
		MCDB	11.9	10.3	13.7	17.7	21.9	26.1	27.2	26.6	25.3	21.8	16.7	14.9
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.1	5.2	4.8	4.0	3.2	3.0	2.5	2.4	3.3	4.0	4.5
		MCDBR	6.7	7.8	7.8	6.8	6.0	4.8	4.3	3.8	3.5	4.1	4.7	5.9
		MCWBR	6.6	7.4	6.1	5.0	4.0	3.5	3.5	3.4	3.3	4.3	5.3	6.1
	5% WB	MCDBR	6.2	8.0	7.8	6.2	4.5	4.4	4.0	3.3	3.1	3.4	4.3	5.7
		MCWBR	6.6	8.0	6.9	5.4	3.8	3.8	3.5	3.5	3.2	4.0	5.4	6.2
Clear-Sky Solar Irradiance	taub	0.316	0.325	0.349	0.396	0.441	0.485	0.514	0.506	0.414	0.360	0.329	0.319	
	taud	2.547	2.516	2.478	2.359	2.267	2.194	2.123	2.132	2.388	2.529	2.558	2.564	
	Ebn at Noon	880	914	917	886	847	807	780	776	838	862	857	852	
	Edh at Noon	77	90	102	121	135	145	155	151	111	88	76	72	
All-Sky Solar Radiation	RadAvg	2.28	3.07	4.17	5.33	6.01	6.46	6.17	5.57	4.57	3.57	2.59	2.04	
	RadStd	0.18	0.31	0.32	0.36	0.50	0.41	0.29	0.35	0.34	0.36	0.20	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+104

Nomenclature: See separate page